LINEAR ALGEBRA · WEEK 3 · SESSION 7

Matrices and determinants

Tables of numbers, the ways to combine them, and the single number that says whether a square matrix squashes space flat.

Matrices, their operations, and determinants up to order 4. Live labs check your work, and the exercises at the end have hints and answers.

1 Matrix → 2 Special matrices → 3 Sum and scalar → 4 Matrix product →
5 Determinant → 6 Adjugates → 7 Properties

Each step uses the one before it. Then: traps, what you need to know, exercises with hints and answers.

STEP 1 · THE OBJECT

Matrix

A table of numbers with a fixed number of rows and columns.

Intuition. A matrix is a table of numbers. You already know one: a grayscale image is a matrix where each element is the brightness of one pixel. The same table can also hold the coefficients of a system of equations, or a set of vectors written as columns.

DEFINITION

A matrix A of dimension $m \times n$ is a table of $m \cdot n$ elements in m rows and n columns:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

a_{ij} is the element in row i and column j. Rows always come first: 2 × 3 means 2 rows and 3 columns.

Example. $A = \begin{pmatrix} 4 & -2 & 7 \\ 0 & 1 & -6 \end{pmatrix}$ has dimension 2 × 3, with $a_{23} = -6$ and $a_{12} = -2$.

Equal matrices. Same dimension and the same element in every position. So an equality of matrices is one equation per position:

$$\begin{pmatrix} a & 1 \\ 4 & b \end{pmatrix} = \begin{pmatrix} 3 & c \\ d & 0 \end{pmatrix} \Leftrightarrow a = 3, b = 0, c = 1, d = 4$$

DEFINITION · TRANSPOSED MATRIX

A^{t} swaps rows and columns: row 1 of A becomes column 1 of A^{t}, and so on. So $(A^{\mathrm{t}})_{ij} = a_{ji}$, an $m \times n$ matrix becomes $n \times m$, and $(A^{\mathrm{t}})^{\mathrm{t}} = A$.

$$A = \begin{pmatrix} 1 & 0 & 4 \\ -2 & 5 & 3 \end{pmatrix} \Rightarrow A^{\mathrm{t}} = \begin{pmatrix} 1 & -2 \\ 0 & 5 \\ 4 & 3 \end{pmatrix}$$

STEP 2 · USES STEP 1

Special matrices

Names for matrices with a recognizable shape. Quizzes ask for them by name.

Main diagonal. In a square matrix, the elements $a_{11}, a_{22}, \ldots, a_{nn}$, from top left to bottom right (shaded below). Only square matrices have one, and most of these names describe what sits on it, above it or below it.

Row matrix

$m = 1$

$$\begin{pmatrix} 2 & 0 & -3 \end{pmatrix}$$

Column matrix

$n = 1$

$$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

Null matrix

Every element is 0. This one is $0_{2\times 3}$.

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Square matrix of order n

$m = n$

$$\begin{pmatrix} 3 & -1 \\ 0 & 2 \end{pmatrix}$$

Symmetric

$A = A^{\mathrm{t}}$, so $a_{ij} = a_{ji}$: a mirror across the diagonal.

$$\begin{pmatrix} 2 & -1 & 5 \\ -1 & 0 & 3 \\ 5 & 3 & 4 \end{pmatrix}$$

Diagonal

$a_{ij} = 0$ whenever $i \neq j$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

Identity I_n

Diagonal, with every $a_{ii} = 1$. It acts like the number 1.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Upper triangular

$a_{ij} = 0$ whenever $i > j$: zeros below the diagonal.

$$\begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{pmatrix}$$

Lower triangular

$a_{ij} = 0$ whenever $i < j$: zeros above the diagonal.

$$\begin{pmatrix} 1 & 0 & 0 \\ -3 & 2 & 0 \\ 4 & 5 & -1 \end{pmatrix}$$

To decode the triangular conditions: $i > j$ means the row number is bigger than the column number, which is below the diagonal.

Labels combine. A diagonal matrix is also symmetric, upper triangular and lower triangular. The identity is all of these at once. Check it below.

MATRIX INSPECTOR · TYPE, DIMENSION AND TRANSPOSE

2 × 3 example | Row matrix | Null 2 × 3 | Symmetric | Diagonal | Identity I_3 | Upper triangular

Rows 2 Columns 3 Type in any box. Shaded boxes: the main diagonal.

4	-2	7
0	1	-6

2 × 3 2 rows, 3 columns

It is: none of the special types.

- ✗ Row matrix · no
  2 rows
- ✗ Column matrix · no
  3 columns
- ✗ Null matrix · no
  $a_{11} = 4$ is not 0
- ✗ Square matrix · no
  2 rows ≠ 3 columns
- ✗ Symmetric · no
  only defined for square matrices
- ✗ Diagonal · no
  only defined for square matrices
- ✗ Identity · no
  only defined for square matrices
- ✗ Upper triangular · no
  only defined for square matrices
- ✗ Lower triangular · no
  only defined for square matrices

$$A^{t} = \begin{pmatrix} 4 & 0 \\ -2 & 1 \\ 7 & -6 \end{pmatrix}$$ 3 × 2

STEP 3 · USES STEP 1

Sum and product by a scalar

Element by element: each position is computed on its own.

DEFINITION · SUM (AND SUBTRACTION)

Only for matrices of the same dimension. Add the elements in the same position: $A + B = (a_{ij} + b_{ij})$.

$$\begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix}$$

Properties: associative, commutative, the null matrix as identity element, and $-A = (-a_{ij})$ as inverse element.

DEFINITION · PRODUCT BY A SCALAR

Multiply every element by the number: $k \cdot A = (k \cdot a_{ij})$.

$$2 \cdot \begin{pmatrix} 3 & -1 \\ 0 & 4 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 6 & -2 \\ 0 & 8 \\ -4 & 2 \end{pmatrix}$$

Properties: $k(A+B) = kA + kB$, $(k+h)A = kA + hA$, $k(hA) = (kh)A$, $1 \cdot A = A$.

STEP 4 · USES STEP 3

Matrix product

The one operation that does not work element by element.

Intuition. Each element of $A \cdot B$ pairs one row of A with one column of B: multiply term by term, then add. Repeat that for every element of the result.

$$(m \times \underline{n}) \cdot (\underline{n} \times p) \to m \times p$$

Inner sizes must match: columns of *A* = rows of *B*. **Outer sizes** give the size of the result.

DEFINITION

If A is $m \times n$ and B is $n \times p$, then $C = A \cdot B$ is $m \times p$ with

$$c_{ij} = a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + \cdots + a_{in} \cdot b_{nj}$$

Worked example. The same two matrices in both orders:

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 + 2 \cdot 2 & 1 \cdot 1 + 2 \cdot 4 \\ 0 \cdot 3 + (-1) \cdot 2 & 0 \cdot 1 + (-1) \cdot 4 \end{pmatrix} = \begin{pmatrix} 7 & 9 \\ -2 & -4 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 2 & 0 \end{pmatrix}$$

Order matters. $A \cdot B \neq B \cdot A$ in general. If A is 2 × 3 and B is 3 × 3, then $A \cdot B$ is 2 × 3, but $B \cdot A$ does not even exist.

PROPERTIES OF THE MATRIX PRODUCT

1. Associative: $(A \cdot B) \cdot C = A \cdot (B \cdot C)$.
2. Not commutative: in general $A \cdot B \neq B \cdot A$. Some pairs commute, like $A \cdot I_n = I_n \cdot A$, but you can never assume it.
3. Identity: for a square A of order n, $A \cdot I_n = I_n \cdot A = A$.
4. Inverse: a matrix B with $A \cdot B = B \cdot A = I_n$ is the inverse A^{-1}. Not every square matrix has one; a matrix without an inverse is singular. Session 8 computes it.
5. Distributive: $A \cdot (B + C) = AB + AC$ and $(A + B) \cdot C = AC + BC$. Keep each matrix on its side.

Extra From properties 2 and 5: $(A + B)^2 = A^2 + AB + BA + B^2$, not $A^2 + 2AB + B^2$. And powers like $A^2 = A \cdot A$ exist only for square matrices.

PRODUCT LAB · ROW TIMES COLUMN

Worked example | 2 × 3 times 3 × 3 | Row times column | Sizes do not match | Times the identity

A rows 2 × columns 2 B rows 2 × columns 2 Swap the order

$$\underset{A}{\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}} \cdot \underset{B}{\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}} = \underset{A \cdot B}{\begin{pmatrix} 7 & 9 \\ -2 & -4 \end{pmatrix}}$$

$(2 \times \underline{2}) \cdot (\underline{2} \times 2) \rightarrow 2 \times 2$

Defined Pick any element of the result to see the row and column it comes from.

c_{11} = row 1 of A · column 1 of B = $1{\cdot}3 + 2{\cdot}2 = \mathbf{7}$

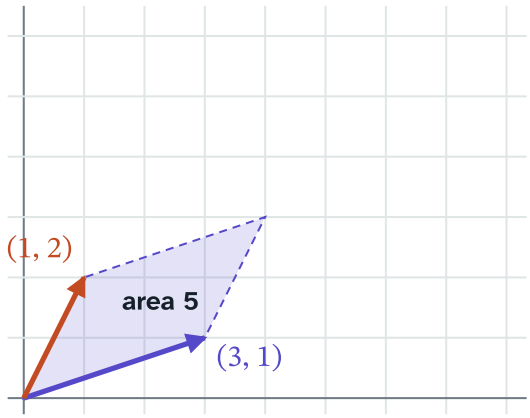
$B \cdot A = \begin{pmatrix} 3 & 5 \\ 2 & 0 \end{pmatrix}$ 2 × 2. Different from A · B: the order matters.

STEP 5 · SQUARE MATRICES ONLY

Determinant of order 2 and 3

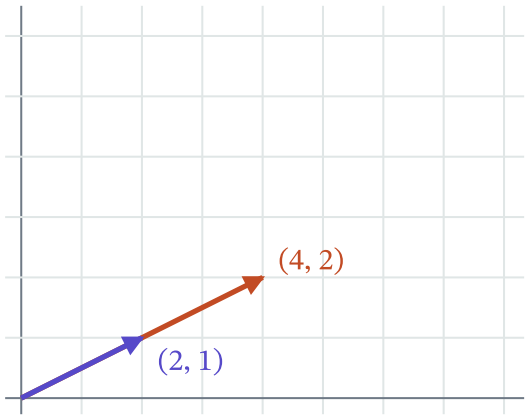
One number per square matrix. It is 0 exactly when the rows are linearly dependent.

Intuition. Extra For order 2, the determinant measures the area of the parallelogram spanned by the two rows, with a sign. If one row is a multiple of the other, the parallelogram collapses into a line and the area is 0. That is why a zero determinant will mean "no inverse" in session 8: a flattened shape cannot be unflattened.



$\det = 3{\cdot}2 - 1{\cdot}1 = 5$

Independent rows



$\det = 2{\cdot}2 - 1{\cdot}4 = 0$

Flat: area 0, dependent rows

Notation. A matrix goes in parentheses and is a table. Its determinant, $\det(A)$ or $|A|$, goes between bars and is one number. For order 1, $|a_{11}| = a_{11}$, so $|-5| = -5$: these bars are not an absolute value.

ORDER 2

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}\cdot a_{22} - a_{12}\cdot a_{21}$$

Main diagonal minus the other diagonal: $\begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 3\cdot 2 - 1\cdot 4 = 2$ and $\begin{vmatrix} -1 & 5 \\ 2 & 3 \end{vmatrix} = (-1)\cdot 3 - 5\cdot 2 = -13.$

ORDER 3 · SARRUS' RULE

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = +\,a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{23}a_{12}$$

$$-\,a_{13}a_{22}a_{31} - a_{23}a_{32}a_{11} - a_{33}a_{21}a_{12}$$

- **Positive:** the main diagonal, plus the two triangles parallel to it.
- **Negative:** the other diagonal, plus the two triangles parallel to it.

Habit that saves points. Always write (positives) − (negatives) with brackets around the negative group. Most Sarrus errors are sign slips inside that group:

$$\begin{vmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ -2 & 1 & 4 \end{vmatrix} = (0 - 3 - 8) - (0 + 2 + 24) = -11 - 26 = -37$$

SARRUS LAB · PICK A PRODUCT TO SEE ITS THREE ELEMENTS

Worked example | Another one | Two equal rows | Identity

POSITIVE

$a_{11}\cdot a_{22}\cdot a_{33} = 1\cdot 0\cdot 4 = 0$

$a_{21}\cdot a_{32}\cdot a_{13} = 3\cdot 1\cdot (-1) = -3$

1	2	-1
3	0	2
-2	1	4

$a_{31}\cdot a_{23}\cdot a_{12} = (-2)\cdot 2\cdot 2 = -8$

NEGATIVE

$a_{13}\cdot a_{22}\cdot a_{31} = (-1)\cdot 0\cdot(-2) = 0$

$a_{23}\cdot a_{32}\cdot a_{11} = 2\cdot 1\cdot 1 = 2$

$a_{\cdot}\cdot a_{\cdot}\cdot a_{\cdot} = 4\cdot 3\cdot 2 = 24$

$$|A| = -37$$

$(0 - 3 - 8) - (0 + 2 + 24) = -11 - 26 = -37$

Selected: a positive product along the main diagonal.

STEP 6 · USES STEP 5

Adjugates and order 4

Break a big determinant into smaller ones, step by step.

Intuition. There is no Sarrus beyond order 3. Instead, pick one row or column and write

DEFINITION · ADJUGATE (OR ADJOINT) OF AN ELEMENT

The adjugate A_{ij} of a_{ij}: delete row i and column j, take the determinant of what remains (the complementary minor), and put the sign $(-1)^{i+j}$ in front.

$$\text{signs:} \begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix} \qquad A_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

The sign depends only on the position: + when $i + j$ is even, – when it is odd. For A_{21}, 2 + 1 = 3 is odd.

EXPANSION ALONG A ROW OR COLUMN

$$|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \quad (\text{row } i)$$

$$|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} \quad (\text{column } j)$$

Every row and every column gives the same result. Check it in the lab below by switching lines.

Choose the line with the most zeros. A zero element cancels its whole term. In a 4 × 4 matrix, a row with three zeros leaves one 3 × 3 determinant instead of four.

Fastest method: make zeros, then expand. Property 8 (step 7) lets you add a multiple of one row to another without changing the determinant:

$$\begin{vmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ -1 & 1 & 2 & 0 \\ 3 & 6 & 1 & 2 \end{vmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 + R_1 \\ R_4 - 3R_1 \\ = \end{matrix} \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 1 & -1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 1 & 1 \\ 3 & 2 & 1 \\ 0 & 1 & -1 \end{vmatrix} = (-2 + 3 + 0) - (0 + 1 -$$

After the row operations, column 1 holds a single non-zero element, $a_{11} = 1$, with sign +. One order-3 determinant is left, solved with Sarrus.

EXPANSION LAB · CHOOSE A LINE, THEN A TERM

4 × 4 with a sparse row | 3 × 3 | After making zeros | Blank 3 × 3 | Blank 4 × 4

+	−	+	−
2	1	0	3
−	+	−	+
0	0	4	0
+	−	+	−
1	-1	2	1
−	+	−	+
3	0	1	2

ADJUGATE OF a_{23}

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 0 & 2 \end{vmatrix}$$

$$= -(6) = \mathbf{-6}$$

Delete row 2 and column 3 (faded), take the determinant of what is left, then apply the sign.

Small signs in the corners: the $(-1)^{i+j}$ chessboard. Dashed boxes: the line you expand along. Faded boxes: the row and column deleted for the selected term.

Row 1 1 zero | Row 2 3 zeros **most zeros** | Row 3 0 zeros | Row 4 1 zero

Column 1 1 zero | Column 2 2 zeros | Column 3 1 zero | Column 4 1 zero

$|A| = -24$ expanding along row 2

$a_{21} \cdot A_{21}$: here $a_{21} = 0$, so the term is 0. Skip it.

$a_{22} \cdot A_{22}$: here $a_{22} = 0$, so the term is 0. Skip it.

$a_{23} \cdot A_{23} = 4 \cdot (-6) = -24$

$a_{24} \cdot A_{24}$: here $a_{24} = 0$, so the term is 0. Skip it.

$|A| = 0 + 0 + (-24) + 0 = \mathbf{-24}$

Choose another row or column: the total never changes.

STEP 7 · USES STEPS 5 AND 6

Properties of determinants

Find a determinant without computing it, or make the computation short.

Numbered for reference. R1 means row 1, C1 means column 1. Summarizing them in your own words is the fastest way to learn them; the last column is a starting point.

#	PROPERTY	TINY EXAMPLE	USE IT TO
1	$\lvert A^t \rvert = \lvert A \rvert$	$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = -2$	Apply every row rule to columns too
2	Swapping two rows (or columns) changes the sign	$\begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix} = 2 = -\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$	Count swaps: each one multiplies by –1
3	Two equal or proportional rows (or columns) give 0	$\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$	Spot a zero at a glance
4	A null row (or column) gives 0	$\begin{vmatrix} 0 & 0 \\ 3 & 4 \end{vmatrix} = 0$	Same

#	PROPERTY	TINY EXAMPLE	USE IT TO
5	Multiplying the determinant by k = multiplying ONE row (or column) by k	$\begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} = 2 \cdot \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 2$	Take a common factor out of a row
6	A row written as a sum splits the determinant in two	$\begin{vmatrix} 1+2 & 2+1 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = -2 + 5$	Only that row splits; the rest stays
7	A row (or column) that is a linear combination of the others gives 0	$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{vmatrix} = 0$, R3 = R1 + R2	Properties 3 and 4 are special cases
8	Adding a linear combination of the OTHER rows to a row changes nothing	$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix}$, R2 − 3R1	Create zeros before expanding
9	$\lvert A \cdot B \rvert = \lvert A \rvert \cdot \lvert B \rvert$	If $\lvert A \rvert = 2$ and $\lvert B \rvert = 3$, then $\lvert A \cdot B \rvert = 6$	Skip computing the product

ROW OPERATIONS AND THEIR EFFECT

OPERATION	EFFECT ON |A|	PROPERTY
$R_i \leftrightarrow R_j$	multiplied by −1	2
$R_i \to k \cdot R_i$	multiplied by k	5
$R_i \to R_i + k \cdot R_j$	unchanged	8

Extra **Consequences worth knowing.** They follow directly from the properties above:

- $\lvert k \cdot A \rvert = k^n \cdot \lvert A \rvert$ for order n, because $k \cdot A$ multiplies all n rows. For a 3 × 3 matrix, $\lvert 2A \rvert = 8\lvert A \rvert$.
- $\lvert A + B \rvert \neq \lvert A \rvert + \lvert B \rvert$ in general. Property 6 splits one row, never the whole matrix.
- Triangular or diagonal matrix: the determinant is the product of the main diagonal. Expand along column 1 again and again to see why.
- Preview of session 8: $\lvert I_n \rvert = 1$, so property 9 gives $\lvert A^{-1} \rvert = 1/\lvert A \rvert$. A matrix with $\lvert A \rvert = 0$ has no inverse.

ROW OPERATIONS LAB · WHAT EACH STEP DOES TO |A|

Challenge matrix | Second challenge | Proportional rows

Challenge: from the first matrix, reach an upper triangular matrix using only "add *k* times another row", then read |*A*| off the diagonal.

Operation: Add k times another row | Row: R_3 | other row: R_1 | k: -5

Apply | Transpose | Undo

Next step: $R_3 \to R_3 - 5 \cdot R_1$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} \qquad |A| = 1$$

Not triangular yet

STEPS, NEWEST FIRST

No steps yet. The suggested first step is ready above.

BEFORE THE EXERCISES

Traps

Seven mistakes that cost the most points.

Common mistakes

1. **Multiplying in the wrong order.** $A \cdot B \neq B \cdot A$. Check the dimensions before multiplying.
2. **Reading |a| as an absolute value.** For a 1 × 1 matrix, $|-5| = -5$.
3. **Sign slips in Sarrus.** Write (positives) – (negatives), brackets included.
4. **Forgetting the adjugate sign.** A_{ij} always carries $(-1)^{i+j}$.
5. **Scaling the row you replace.** $R_2 \to 2R_2 - R_1$ multiplies the determinant by 2. Write every step as $R_i \to R_i + k \cdot R_j$. Scaling the replaced row is fine when you

solve a system of equations, but not when you compute a determinant.

6. **Mixing up $k\cdot|A|$ and $|k\cdot A|$.** The first scales one row, the second all of them: $k^n\cdot|A|$.
7. **Splitting a whole sum.** $|A+B|$ is not $|A|+|B|$.

Terminology. This page calls A_{ij} the adjugate (or adjoint) of an element, and $\mathrm{adj}(A)$ the matrix of adjugates. Many textbooks and videos call A_{ij} the cofactor and use "adjugate" for the already transposed matrix. Check which convention your course uses, so you never transpose twice.

REVIEW

What you need to know

Tick items off as you master them. Your ticks stay in this browser.

0 of 22 checked Clear ticks

Definitions to state

- [ ] Matrix of dimension $m \times n$ and element a_{ij}
- [ ] Equal matrices
- [ ] Transposed matrix A^{t}
- [ ] Row, column, null and square matrices; main diagonal
- [ ] Symmetric, diagonal, identity, upper and lower triangular
- [ ] Inverse matrix and singular matrix
- [ ] Determinant $|A|$ of order 1, 2 and 3
- [ ] Adjugate A_{ij} and complementary minor

Skills the exercises test

- [ ] Add, subtract and scale matrices
- [ ] Decide whether a product exists and give its size
- [ ] Multiply 2 × 2 and 3 × 3 matrices by hand
- [ ] Compute determinants of order 2 and 3 (Sarrus)
- [ ] Compute an adjugate with its correct sign
- [ ] Make zeros with row operations, then expand an order-4 determinant
- [ ] Spot a zero determinant from the properties alone

Facts to memorize

- [ ] $(m \times n)\cdot(n \times p) \to m \times p$
- [ ] $A \cdot B \neq B \cdot A$ in general
- [ ] $(A + B)^2 = A^2 + AB + BA + B^2$
- [ ] The sign of A_{ij} is $(-1)^{i+j}$
- [ ] The nine properties, by number
- [ ] Swap: × (–1). Scale a row: × k. Add a multiple: no change
- [ ] $|AB| = |A||B|$, but $|A + B| \neq |A| + |B|$

PRACTICE · AFTER THE SESSION

Exercises

Do them on paper in a separate sitting. Open a hint only when stuck, and the answer only when done.

A · Matrices

A1 Give the dimension of A, the elements a_{13} and a_{22}, and A^{t} with its dimension. Step 1

$$A = \begin{pmatrix} 5 & 0 & -2 \\ 1 & -3 & 4 \end{pmatrix}$$

Show hint

Show answer

A2 Classify each matrix: row, column, null, square, symmetric, diagonal, identity, upper or lower triangular. More than one label can apply. Step 2

$$M1 = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix} \quad M2 = \begin{pmatrix} 1 & 4 \\ 4 & 0 \end{pmatrix} \quad M3 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix} \quad M4 = \begin{pmatrix} 5 & -1 & 0 \end{pmatrix}$$

Show hint

Show answer

A3 Find a, b, c and d so that the matrices are equal. Step 1

$$\begin{pmatrix} a & 3 \\ 2b & c \end{pmatrix} = \begin{pmatrix} 5 & d-1 \\ 4 & 0 \end{pmatrix}$$

Show hint

Show answer

A4 Compute $2A - B$, $A \cdot B$, $B \cdot A$ and $A \cdot C$. Does $C \cdot A$ exist? Steps 3 and 4

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix} \quad C = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Show hint

Show answer

A5 With A and B from A4, compute $(A + B)^2$ and $A^2 + 2AB + B^2$. Why are they different? Step 4

Show hint

Show answer

A6 Using only the definition of the inverse, check that B is the inverse of A. Step 4

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$$

Show hint

Show answer

B · Determinants

B1 Compute both. Simplify b) completely. Step 5

a) $\begin{vmatrix} 4 & -3 \\ 2 & 5 \end{vmatrix}$ b) $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix}$

Show hint

Show answer

B2 Compute with Sarrus' rule. Step 5

$$\begin{vmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \\ 4 & 1 & -1 \end{vmatrix}$$

Show hint

Show answer

B3 Compute the same determinant again by expanding along column 2. Your result must match B2. Step 6

Show hint

Show answer

B4 For the matrix in B2, compute the adjugates A_{12}, A_{31} and A_{23}. Step 6

Show hint

Show answer

B5 Compute. Step 6

$$\begin{vmatrix} 2 & 0 & 1 & 3 \\ 1 & 0 & 0 & 2 \\ -1 & 3 & 2 & 1 \\ 0 & 0 & 4 & 1 \end{vmatrix}$$

Show hint

Show answer

B6 Compute, making zeros in column 1 first with steps of the form $R_i \to R_i + k \cdot R_1$. Steps 6 and 7

$$\begin{vmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 1 \\ -1 & 0 & 2 & 2 \\ 3 & 7 & 4 & 2 \end{vmatrix}$$

Show hint

Show answer

C · Properties

C1 Without computing, explain why each determinant is 0 and name the property. Step 7

a) $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{vmatrix}$ b) $\begin{vmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 7 & 0 & 1 \end{vmatrix}$ c) $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{vmatrix}$ d) $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 8 & 0 \\ 2 & 4 & 5 \end{vmatrix}$

Show hint

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C2 A is 3 × 3 with rows (a, b, c), (d, e, f), (g, h, i) and $|A| = 4$. Find each value and name the property. Step 7

a) $\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix}$ b) $\begin{vmatrix} 2a & 2b & 2c \\ d & e & f \\ g & h & i \end{vmatrix}$ c) $\begin{vmatrix} a & b & c \\ d+3a & e+3b & f+3c \\ g & h & i \end{vmatrix}$

d) $|A^t|$ e) $|A \cdot A|$ f) $|2A|$

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C3 Make zeros with property 8, then compute. Step 7

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{vmatrix}$$

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C4 Check property 9 by computing $|A|$, $|B|$ and $|A \cdot B|$. Then show that $|A + B| \neq |A| + |B|$. Steps 4 and 7

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$$

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C5 Exam level. Find every real x for which this determinant is 0. Steps 5 and 7

$$\begin{vmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{vmatrix}$$

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Items marked Extra go beyond the core topics but help in exercises.

Every answer on this page was checked by computer. Still check your own work with your math software.